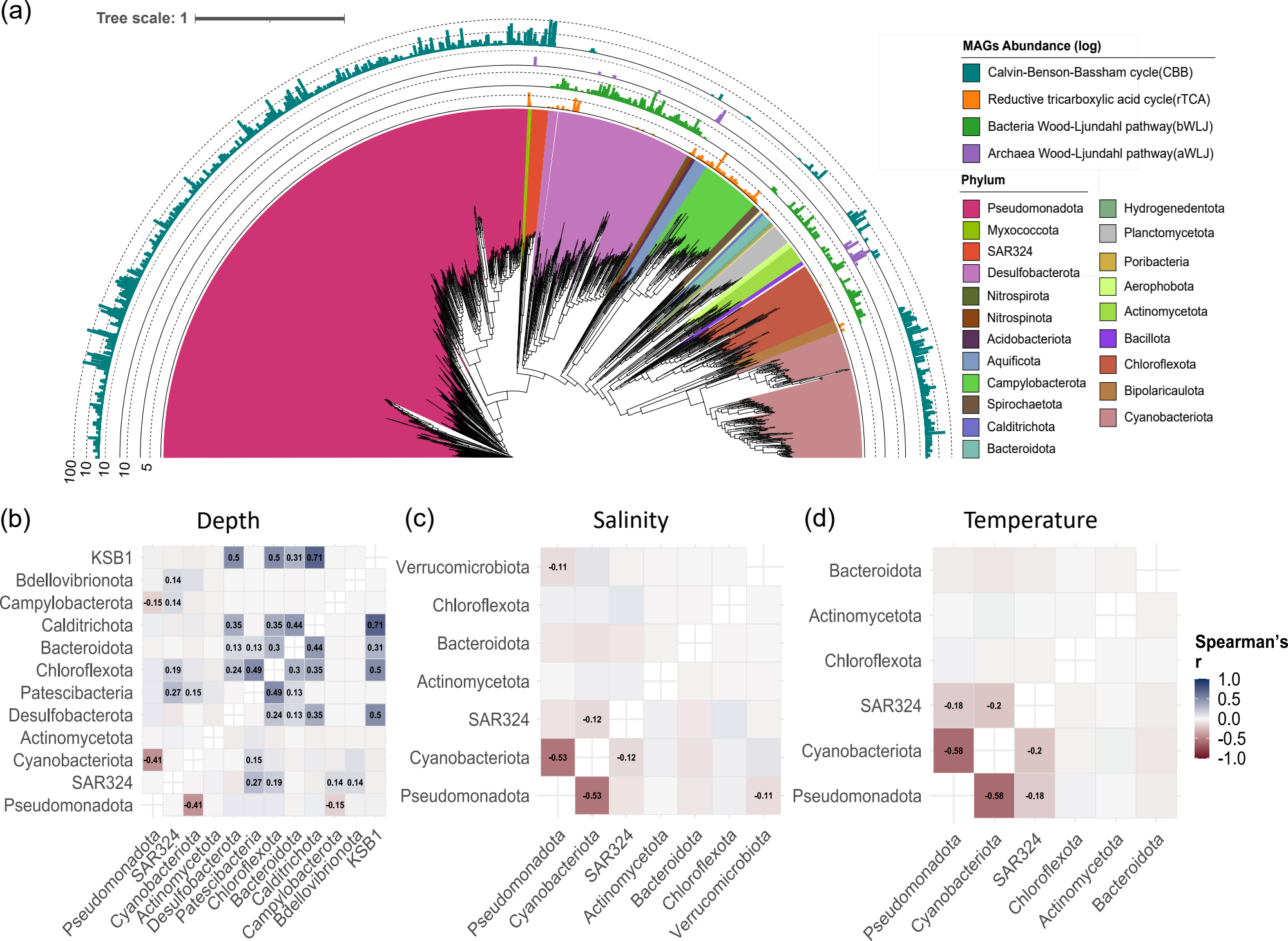
(a)
Tree scale: 1
MAGs Abundance (log)
Calvin-Benson-Bassham cycle(CBB)
Reductive tricarboxylic acid cycle(rTCA)
Bacteria Wood-Ljundahl pathway(bWLJ)
Archaea Wood-Ljundahl pathway(aWLJ)
Phylum
Pseudomonadota
Myxococcota
SAR324
Desulfobacterota
Nitrospirota
Nitrospinota
Acidobacteriota
Aquificota
Campylobacterota
Spirochaetota
Calditrichota
Bacteroidota
Hydrogenedentota
Planctomycetota
Poribacteria
Aerophobota
Actinomycetota
Bacillota
Chloroflexota
Bipolaricaulota
Cyanobacteriota
100
10
10
10
5
(b)
Depth
KSB1
Bdellovibrionota
Campylobacterota
Calditrichota
Bacteroidota
Chloroflexota
Patescibacteria
Desulfobacterota
Actinomycetota
Cyanobacteriota
SAR324
Pseudomonadota
Pseudomonadota
SAR324
Cyanobacteriota
Actinomycetota
Desulfobacterota
Patescibacteria
Chloroflexota
Bacteroidota
Calditrichota
Campylobacterota
Bdellovibrionota
KSB1
(c)
Salinity
Verrucomicrobiota
Chloroflexota
Bacteroidota
Actinomycetota
SAR324
Cyanobacteriota
Pseudomonadota
Pseudomonadota
Cyanobacteriota
SAR324
Actinomycetota
Bacteroidota
Chloroflexota
Verrucomicrobiota
(d)
Temperature
Bacteroidota
Actinomycetota
Chloroflexota
SAR324
Cyanobacteriota
Pseudomonadota
Pseudomonadota
Cyanobacteriota
SAR324
Chloroflexota
Actinomycetota
Bacteroidota
Spearman's r
1.0
0.5
0.0
-0.5
-1.0